



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to TAMILARASI FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

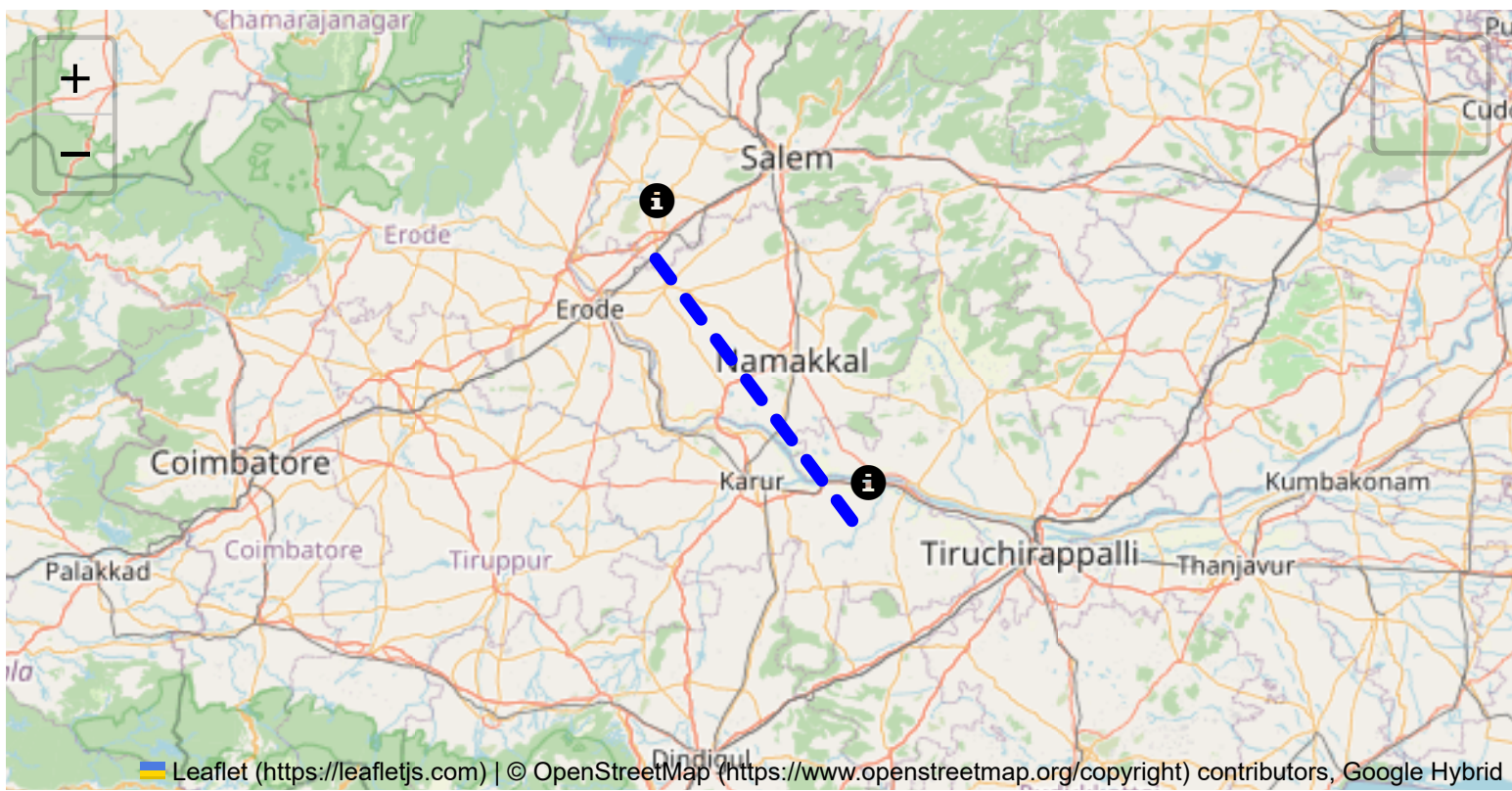
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 110.02 km
Estimated Duration: 2.2 hours
Adjusted Duration (Heavy Vehicle): 2.7 hours
Start: (11.4381, 77.8734)
End: (10.835309, 78.332909)



Welcome to the Journey Risk Management Study

Route Overview

The route from Sangagiri, Tamil Nadu to Pothuravuthanpatti, Tamil Nadu stretches approximately 110.02 kilometers and typically takes about 2.19 hours for heavy vehicles carrying hazardous materials. The journey includes passage through state highways and rural roads, which can fluctuate in complexity and traffic density.

Weather Conditions

Typical weather in Tamil Nadu includes a tropical climate with hot summers, a significant monsoon season typically from June to September, and mild winters. Key weather hazards can include heavy rainfall during the monsoon, leading to flooding or reduced visibility, and extreme heat during the summer months, which can affect vehicle performance and road conditions.

Traffic Patterns

Traffic tends to peak during morning (8 AM - 10 AM) and evening hours (5 PM - 7 PM). Certain segments, particularly near Sangagiri and larger towns along the route, may experience congestion. Rural segments might be less trafficked but watch out for local traffic, especially near agricultural areas where slow-moving farm vehicles may be present.

Road Quality and Infrastructure

The route includes a mix of well-maintained state highways and narrower rural roads. Road quality may vary significantly, with potential for potholes, uneven surfaces, and narrow lanes particularly in rural areas. Bridges and culverts should be navigated with caution, especially during or after heavy rains.

Alternative Routes

In case of emergencies, consider using National Highway 44 as an alternative, intersecting with other state highways that provide a bypass around major towns. It's important to stay updated on roadwork and potential closures by consulting local traffic advisories.

Local Regulations

Ensure compliance with Tamil Nadu's regulations on the transport of hazardous materials, including proper labeling, documentation, and safety equipment. It's vital to respect environmental protection rules, especially near agricultural lands and sensitive ecosystems.

Historical Incidents

There are occasional incidents involving heavy vehicles on this route. Known issues often arise from vehicle overturns due to speeding or miscalculating road conditions, particularly under wet conditions. It's advisable to check local news updates for specific incident reports.

Environmental Considerations

The route passes through agricultural areas and proximity to the Kaveri River in parts. Spills or accidents involving hazardous materials could pose severe risks to the environment and local crops. Drivers should be trained to manage incidents without impacting these sensitive areas.

Communication Coverage

Most of the route has decent mobile network coverage, but rural sections may have reduced signal strength leading to potential dead zones. Secondary communication devices should be considered as a backup.

Emergency Response Times

Emergency response times can vary greatly, estimated between 20 to 50 minutes depending on proximity to larger towns with emergency services. It is advisable to have a list of local contact numbers for first responders in each major segment of the route.

Overall Summary

This route, while common for goods transport, presents risks associated with variable road conditions, weather impacts, and traffic congestion. Preparedness for emergencies, compliance with local regulations, and awareness of environmental and communication limitations are crucial for safe transportation of hazardous materials. Continuous monitoring of weather forecasts, road conditions, and traffic updates will help mitigate risks associated with this journey.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Roundabout	High	10.92958, 78.05452	15 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44019, 77.87547	15 KM/Hr
4	Turn	Medium	11.39685, 77.88861	30 KM/Hr
5	Turn	High	11.37862, 77.89488	15 KM/Hr
6	Turn	High	11.37850, 77.89453	15 KM/Hr
7	Turn	Medium	11.16950, 78.01544	30 KM/Hr
8	Turn	Medium	11.16764, 78.01565	30 KM/Hr
9	Turn	High	11.14545, 78.02208	15 KM/Hr
10	Turn	High	11.14480, 78.02291	15 KM/Hr
11	Turn	Medium	11.10102, 78.00572	30 KM/Hr
12	Turn	Medium	11.05261, 78.05138	30 KM/Hr
13	Turn	High	10.93631, 78.22884	15 KM/Hr
14	Turn	Medium	10.93602, 78.22945	30 KM/Hr
15	Turn	Medium	10.93152, 78.23153	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
16	Turn	Medium	10.92819, 78.23743	30 KM/Hr
17	Turn	High	10.91798, 78.23849	15 KM/Hr
18	Turn	Medium	10.91738, 78.23888	30 KM/Hr
19	Turn	Medium	10.90078, 78.23917	30 KM/Hr
20	Turn	High	10.89479, 78.24318	15 KM/Hr
21	Turn	Medium	10.89054, 78.24320	30 KM/Hr
22	Turn	Medium	10.88989, 78.24281	30 KM/Hr
23	Turn	High	10.88000, 78.24306	15 KM/Hr
24	Turn	High	10.87955, 78.24645	15 KM/Hr
25	Turn	Medium	10.87709, 78.24676	30 KM/Hr
26	Turn	Medium	10.87552, 78.24781	30 KM/Hr
27	Turn	High	10.86831, 78.24800	15 KM/Hr
28	Turn	High	10.86764, 78.25008	15 KM/Hr
29	Turn	High	10.86378, 78.24971	15 KM/Hr
30	Turn	High	10.86348, 78.25042	15 KM/Hr
31	Turn	Medium	10.85148, 78.25293	30 KM/Hr
32	Turn	High	10.84470, 78.25878	15 KM/Hr
33	Turn	Medium	10.84294, 78.25857	30 KM/Hr
34	Turn	High	10.84262, 78.25876	15 KM/Hr
35	Turn	High	10.83209, 78.27386	15 KM/Hr
36	Turn	Medium	10.83268, 78.27737	30 KM/Hr
37	Turn	Medium	10.82937, 78.30703	30 KM/Hr
38	Turn	High	10.82922, 78.30825	15 KM/Hr
39	Turn	High	10.83059, 78.30942	15 KM/Hr
40	Turn	Medium	10.83149, 78.32134	30 KM/Hr
41	Turn	Medium	10.83174, 78.32242	30 KM/Hr
42	Turn	High	10.83317, 78.32326	15 KM/Hr
43	Turn	Medium	10.83451, 78.32852	30 KM/Hr
44	Turn	Medium	10.83567, 78.32940	30 KM/Hr
45	Turn	Medium	10.83598, 78.33039	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium
19	hospital	Panjapatti Government Hospital	10.83121, 78.310053	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium
18	college	valluvar catering	10.9553473, 78.0606583	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr
Coordinates: 10.92958, 78.05452



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44019, 77.87547



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39685, 77.88861



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37862, 77.89488



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37850, 77.89453



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.16950, 78.01544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.16764, 78.01565



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.14545, 78.02208



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.14480, 78.02291



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.10102, 78.00572



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.05261, 78.05138



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.93631, 78.22884



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.93602, 78.22945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.93152, 78.23153



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.92819, 78.23743



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.91798, 78.23849



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.91738, 78.23888



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.90078, 78.23917



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.89479, 78.24318



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.89054, 78.24320



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.88989, 78.24281



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.88000, 78.24306



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.87955, 78.24645



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.87709, 78.24676



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.87552, 78.24781



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.86831, 78.24800



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.86764, 78.25008



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.86378, 78.24971



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.86348, 78.25042

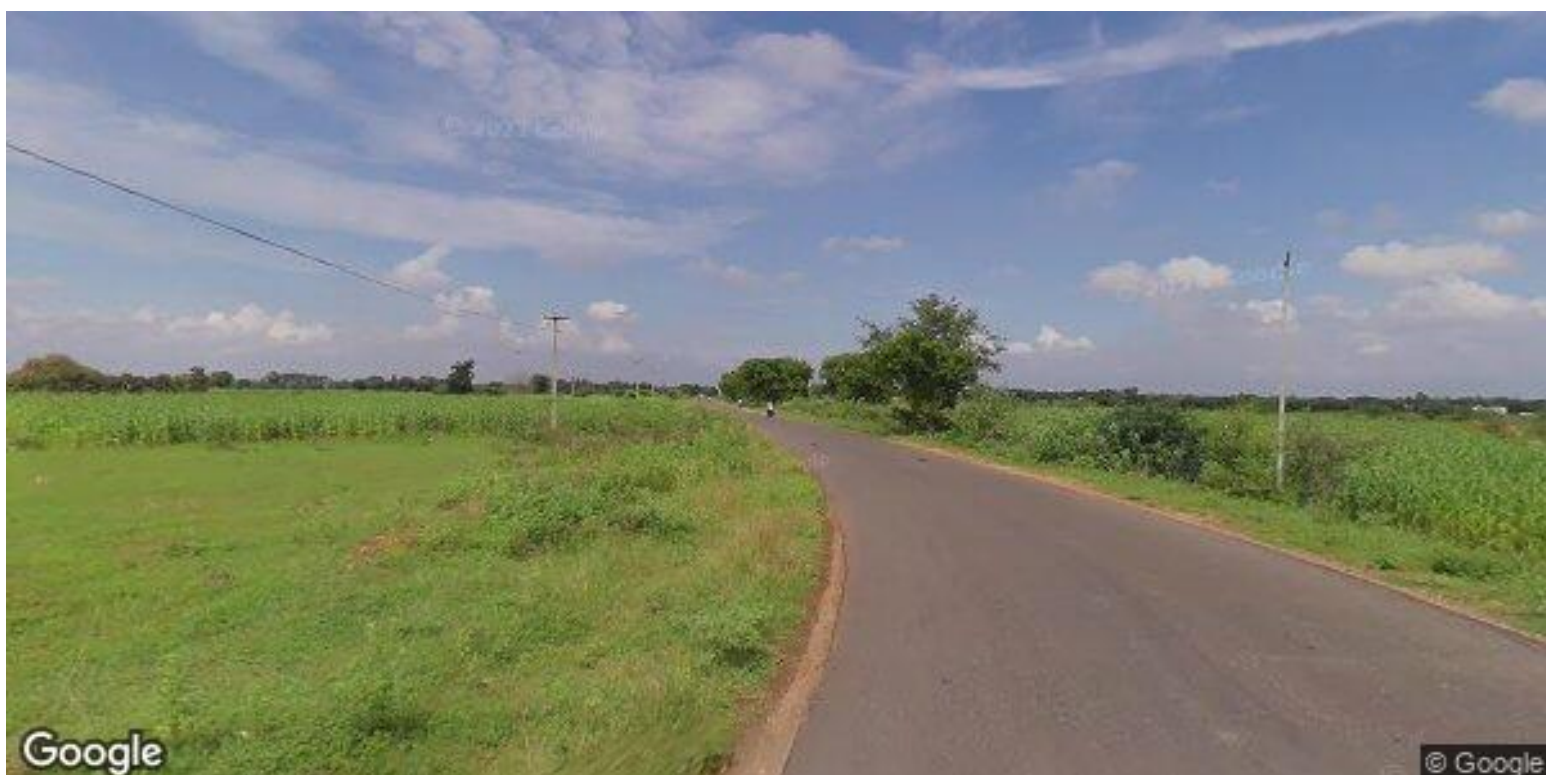


Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.85148, 78.25293



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.84470, 78.25878



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.84294, 78.25857



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.84262, 78.25876



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 10.83209, 78.27386



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.83268, 78.27737



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.82937, 78.30703



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.82922, 78.30825



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.83059, 78.30942



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.83149, 78.32134



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.83174, 78.32242



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 10.83317, 78.32326



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.83451, 78.32852



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 10.83567, 78.32940



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 10.83598, 78.33039
